# Krutarth Parmar

Computer Vision Engineer | Gujarat, India

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## PROFESSIONAL SUMMARY

Computer Vision Engineer with hands-on expertise in YOLO, OpenCV, and PyTorch for detection, segmentation, and tracking tasks. Delivered production CV systems for Indian manufacturing and retail companies. Skilled in model optimisation for edge deployment.

## TECHNICAL SKILLS

**Data Engineering:** Kafka, SQL

**ML Libraries & Frameworks:** Scikit-learn, Pandas, Keras, PyTorch, NumPy, LightGBM

**Programming Languages:** R, Java, Python

**Tools & Technologies:** VS Code, Jupyter, Pandas, NumPy, GitHub

**Soft Skills:** Project Management, Analytical Thinking, Team Collaboration, Leadership

## PROJECTS

### Medical Image Classification using CNN

*Tech Stack: NumPy, Python*

- Built a ResNet-50 transfer learning model for pneumonia detection from chest X-rays.
- Trained on NIH dataset (10K images), achieving 94% accuracy and 0.92 AUC.
- Implemented Grad-CAM heatmaps for model interpretability and clinical validation.

### Sentiment Analysis Engine for E-commerce Reviews

*Tech Stack: Python, Pandas*

- Fine-tuned a BERT-based model on 200K e-commerce reviews achieving 91% F1 score.
- Built an ETL pipeline to ingest reviews daily from multiple platforms using Airflow.
- Deployed the model as a REST API serving 5,000+ inference requests per day.

### Real-Time Object Detection for Smart Surveillance

*Tech Stack: Python, PyTorch*

- Trained a YOLOv8 model on a custom 15K-image dataset for multi-class object detection.
- Achieved mAP@50 of 0.87 with real-time inference at 30 FPS on edge devices.
- Integrated OpenCV pipeline for video stream processing and alert generation.

## EDUCATION

### B.Tech – Artificial Intelligence & Machine Learning

*Caltech CTME | 2024 | CGPA: 10.0*

### Class XII – Kendriya Vidhyalaya, CBSE

*2018 | 68.0%*

### Class X – Kendriya Vidhyalaya, CBSE

*2014 | 80.0%*

## CERTIFICATIONS

- IBM Data Science Professional Certificate